

GEP Intern – Technology: Question Practice Roadmap

FINAL — do not revise; only tick and practice

Ojas | Companion: [GEP_Prep_Roadmap.pdf](#)

Sudhanva BM + LeetCode 2026 + Harsh/Sarthak/GFG. iMocha has no full free public paper.

0) Anchors (read once, then practice)

- [LeetCode — GEP Summer Intern 2026](#)
- Sudhanva: AI MCQs major; OS/DBMS; Easy strings/hashing; JS fill-blanks; R1 named/Easy DSA + dry-run; R2 project depth
- Sample OA codes: `abcde+10101→AbCdE`; CSV max; **Kadane** (and others by name)

1) OA blocks → links

Block	Do
AI MCQs (MAJOR)	Link A — once; intro only
OS / DBMS MCQs	Links F, G
DSA Easy coding	Sec 3–4
JS fill-in-blanks	Link B + Sec 5
Aptitude / soft (if any)	Links C, D, E

A AI aptitude — 25 MCQs w/ answers

B W3Schools JS Quiz

C [How to take iMocha](#)

D IndiaBix — timed 15–20 Q sets

E Free SJT (teamwork / deadlines / honesty)

F Examveda OS MCQs + GFG OS Interview Qs

G IndiaBIX Database MCQs + GFG DBMS Interview Qs

Also: [W3 SQL Quiz](#) | [OOP Concepts quiz](#) | [OOP Interview Qs](#)

AI intro checklist (no math spiral)

- ☐ AI vs ML vs DL ☐ Supervised vs unsupervised idea
- ☐ Linear regression — what it predicts (not derive)
- ☐ Where **CV** is used ☐ Where **RL** is used
- ☐ NLP 1-liner ☐ Data / bias awareness

2) Core CS

- ☐ F + G quizzes ☐ GFG OOP Qs + OOP Concepts quiz
- ☐ SQL: INNER/LEFT JOIN; books by author X by hand

3) Named algorithms (ask-by-name ready)

*Each: 30s idea + when to use + dry-run/code if asked. **Not Kadane-only.***

- ☐ Kadane (max subarray) ☐ Binary search
- ☐ Two pointers / sliding window ☐ BFS / DFS (level-order = BFS)
- ☐ Dijkstra (shortest path idea) ☐ Merge / Quick / Heap sort (time, stable?)
- ☐ Prefix sums ☐ Hashing (Two Sum / freq map)
- ☐ Light DP names: Fib / knapsack 1-liner
- Skip unless on resume: B-Tree / Red-Black internals

4) Coding list

One language. Speak + dry-run. Map to named algos above.

- ☐ Maximum Subarray (Kadane) ☐ Binary Search template
- ☐ Case-map string ☐ First unique char ☐ Two Sum
- ☐ Sliding window Easy ☐ Valid parentheses
- ☐ Level-order (BFS) ☐ Number of islands (DFS/BFS)
- STL / basics: **vector**, **map**/**unordered_map**, **queue**, **string**
- Optional coverage: NeetCode/LC 150 Easy–Medium filter; light Striver — not endless grind

5) JS fill-in

- ☐ Link B once
- ☐ Snippet drills: `==/===`, loops, **map/filter**, string methods — fill only what is asked

6) Interview bank

- ☐ “Explain X algorithm” for every name in Sec 3
- ☐ Kadane dry-run on `[-2,1,-3,4,-1,2,1,-5,4]`
- ☐ Projects: IFFCO / LogiFlow(+SSR) / Community Hero / AirHelp — role, impact, 2–3 follow-ups deep
- ☐ OOP pillars; Stack vs Queue; Mongo vs Postgres; open to Angular/.NET?
- ☐ End Q for them: scale / AI agents / something real
- ☐ HR: layman pitch; confident replies; location #1

7) Execute order (tick down; stop when done bar green)

1. ☐ A (AI) once
2. ☐ F + G (OS/DBMS)
3. ☐ Sec 3 named list — speak each 30s
4. ☐ Code Kadane + Binary Search + BFS + dry runs
5. ☐ Four more Easy from Sec 4
6. ☐ B (JS) + one fill-in snippet
7. ☐ OOP quiz + SQL JOIN

8. ☐ Project depth 20 min aloud
9. ☐ Optional C aptitude samples
10. ☐ One mini mock sitting (AI + Easy + JS blank)

8) Done bar (freeze when all checked)

- ☐ A, F, G ☐ Named list spoken ☐ Kadane+BS+BFS coded
- ☐ ≥ 5 Easy ticks ☐ JS fill-in ☐ Projects deep ☐ Mini mock

No more doc updates. Only practice. Strategy stays in `GEP_Prep_Roadmap.pdf`.